Indicative price for a 10-year solar PPA (Borrentin, 114 MWp)

Memo · 03 Nov 2026 · ppa-lab 1.16.0 · illustrative, from public data: not an offer, no view on actual contracts · details: Step 7 report

Bottom line. For a 10-year pay-as-produced PPA (2027-2036) the negotiation range is **22.6 to 37.2 EUR/MWh**: from the indicative price a buyer can accept after its risk premium and costs, up to the fair value before risk. At the indicative price the PPA cuts our 95% cash flow at risk from EUR 3.24 m to 0.79 m per year (**–76%**) for about **15.0 EUR/MWh** (EUR 1.81 m per year) against selling at spot.

The price, layer by layer (EUR/MWh)

Layer	Solar park	Wind park
Forward 2027 (average of the scenarios)	155.3	155.3
Shape: output falls in cheap hours	–67.5	–11.7
= Fair price 2027	**87.8**	**143.7**
Later years: lower forwards and capture rates; degradation	–50.6	–54.0
= Fair price, 10 years	**37.2**	**89.7**
Buyer's risk premium after a baseload hedge	–11.4	–5.6
Balancing and route to market (assumed)	–2.2	–1.6
Credit and liquidity (assumed)	–1.0	–1.0
= Indicative price, 10 years	**22.6**	**81.5**

What we keep and what we give away

- **Volume** stays with us: P50 120.9 GWh a year, one year in ten below 113.4 GWh (P90, 31 weather years).
- **Price and shape** go to the buyer. Its premium is mostly profile risk for solar (how deep the midday dip is) and volume risk for wind; the price level it can hedge with forwards.
- Against spot sales we give away the premium, the credit cost and the 2.6 EUR/MWh that stopping at negative prices saves (balancing is paid either way).

What would change the price

- Build-out scenarios for solar and storage after 2027: **–3.5 / +8.8**; ±1 pp a year of trend on top: **±3.5** EUR/MWh
- Forward curve –10% / +10%: **–5.8 / +5.7 EUR/MWh**
- Capture rate as far off as the model was out of sample in earlier years (+0.03, Step 6): **–2.2 EUR/MWh**

For comparison, the risk premium and the costs together take 14.6 EUR/MWh off the fair price.

Recommendations

1. Quote within the range above; do not go below 22.6 EUR/MWh.
2. Offer a negative-price clause as the alternative: we stop producing at negative prices (15% of the solar output in 2027, more later) and are paid 41.0 EUR/MWh per delivered MWh: 28.8 per MWh of output against 22.6 without the clause, at a higher cash flow at risk (EUR 1.35 m instead of 0.79 m).
3. Agree the cannibalisation view with Risk before signing, and for any unsold volume sell about 84% of the average output as Cal base futures: it cuts the merchant cash flow at risk by 72% (solar park).

Method and limits

496 joint scenarios for 2027: 31 ERA5 weather years at the real sites, each with simulated hourly prices that respond to that weather (calibrated on 2024-2026 German day-ahead prices, analog days from 2025-10-01). Fair price = E[Σ q P] / E[Σ q]; premium = 5% shortfall of the buyer's hedged value per MWh; later years by the forward curve, capture rates moving with the price level as in the model (and 2023-2026 history), solar's with the build-out, discounted at 6%. Forward: EEX German Power Cal-27 base 154.42 EUR/MWh (quote of 2026-10-23, via tacto.ai); its quarters follow the market's seasonal shape (Q1-Q4 198.4 / 128.8 / 137.9 / 153.2, quotes of 2026-10-04); later years follow the forward curve (2028: 113.94, 2029: 87.93), held flat after 2029. Limits: the forwards are published quotes from secondary sources, not EEX files, the costs are illustrative, grid curtailment is not modelled, and the shape comes from one market regime (2024-2026). The Cal-27 forward moved between 104 and 154 EUR/MWh since 2026-07-31 (published quotes): refresh the offer with the forward curve on the day of signing.